

NIIS HACKATHON 2026 — OFFICIAL PROBLEM BRIEF

Track: Campus Nexus

Challenge ID: PS-01

Title: AI-Powered Campus Life Management Platform

1. Executive Summary & Problem Overview

Modern educational campuses manage hundreds of daily administrative and student operations across disconnected channels—physical logbooks, manual approval slips, scattered messaging apps, and standalone portals. This fragmentation creates significant operational latency, security blind spots at campus checkpoints, unmonitored grievance escalations, and administrative overhead.

The objective of this challenge is to build a centralized, secure, multi-role digital ecosystem powered by AI that unifies student lifecycle interactions, gate permissions, hostel and dining logistics, automated certification, and actionable institutional intelligence into a single accessible portal.

2. Core Operational Scope & Target Personas

A. User Roles & Access Hierarchy

1. **Students:** View personalized schedules, log attendance anomalies, apply for digital out-passes, manage mess services, report grievances, and request verifiable credentials.
2. **Faculty & Mentors:** Track lecture attendance, broadcast class notices, review academic leave requests, and monitor student academic health.
3. **Campus Administration & Wardens:** Review automated gate-pass approvals, manage room allocations, track mess inventory, and broadcast institutional alerts.
4. **Security Staff (Guard Gate Client):** High-speed optical QR/barcode scanner to validate gate-passes in real-time with offline cache redundancy.

3. Key Functional Modules & Technical Deliverables

Module 1: Smart Attendance & Dynamic Timetable

- Real-time automated timetable sync with room and faculty scheduling.
- Geofenced or biometric/credential-backed attendance logging with fraud prevention.

- Predictive absenteeism alerts flagging students falling below institutional thresholds (e.g., $<75\%$).

Module 2: Digital Out-Pass & Automated Gate Security

- Multi-tiered out-pass workflow: Student requests pass $\rightarrow$ Parental/Warden consent/auto-rule evaluation $\rightarrow$ Cryptographically signed dynamic QR pass generated.
- Guard check-in/check-out scanner verifying pass validity, entry/exit timestamps, and curfew status.
- Emergency panic escalation and unauthorized absence alerts.

Module 3: Hostel & Dining (Mess) Intelligence

- Digital mess rebate system allowing students to notify meal opt-outs in advance.
- Predictive food wastage analytics dashboard for mess supervisors based on student headcount.
- Hostel maintenance ticketing (plumbing, electrical, hygiene) with SLA tracking.

Module 4: Digital Certificate Request & Verification

- Automated self-service issuance for Bonafide, NOC, Gate Pass history, and Fee certificates.
- Verifiable digital signature / QR verification code on exported PDF credentials.

Module 5: AI-Driven Grievance & Institutional Intelligence

- NLP-driven complaint routing: Automatic categorization (Ragging, Academics, Infrastructure, Mess) and priority scoring.
- Sentiment analysis on campus feedback to identify systemic operational bottlenecks.
- Unified administrative dashboard displaying live campus health metrics.

4. Recommended Technology Stack

Layer, Suggested Technologies

Frontend Web, "React.js (Vite), Next.js, or Vue.js with Tailwind CSS"

Mobile / Scanner App, "React Native, Flutter, or Progressive Web App (PWA) with HTML5 Camera API"

Backend & APIs, "Node.js (Express), Python (FastAPI/Django), or Go"

Database, PostgreSQL / MongoDB / Firebase Firestore

AI / Machine Learning, OpenAI API / Gemini API / LangChain / HuggingFace (Transformers for categorization & summarization)

Authentication & Security, "JWT, OAuth2, Role-Based Access Control (RBAC), and QR hashing (SHA-256)"

5. Evaluation & Judging Criteria (100 Points Total)

1. **System Architecture & Completeness (25 Points):** Cohesion between multiple modules (Passes, Attendance, Mess, Complaints) in a single platform.
2. **AI & Automation Utility (20 Points):** Real utility of AI implementation (NLP ticket routing, predictive meal planning, anomaly detection) rather than superficial wrappers.
3. **Security & Role-Based Integrity (20 Points):** Protection of student records, verification of gate QR codes against tampering, and role segregation.
4. **UI/UX & Operational Efficiency (20 Points):** Mobile-first responsiveness, minimal steps for critical actions (e.g., sub-2-second gate check-in).
5. **Pitch, Code Quality & 24h Sprint Execution (15 Points):** Clean repo structure, working local prototype demonstration, and edge-case handling.

6. Submission Guidelines

- **Repository:** Public/Private GitHub repository with structured commits, .env.example, and clear README.md setup instructions.
- **Live Deployment:** Working live deployment link (Vercel / Render / Firebase / Netlify).
- **Demo Video / Deck:** Maximum 3-minute video overview demonstrating the multi-role workflow.